Supplemental Material

A Multilevel Signal Detection Theory Approach to Unconscious Processing Research

Ricardo Rey-Sáez[1], Alicia Franco-Martínez[1], Miguel A. Vadillo[2], and Francisco Garre-Frutos[1]

[1]Department of Basic Psychology, Autonomous University of Madrid

[2]Department of Basic Psychology I, Universidad Nacional de Educación a Distancia

Author Note

Ricardo Rey-Sáez https://orcid.org/0000-0001-6739-2035

Alicia Franco-Martínez https://orcid.org/0000-0002-9710-1240

Miguel A. Vadillo https://orcid.org/0000-0001-8421-816X

Francisco Garre-Frutos https://orcid.org/0000-0001-9810-186X

All data, models, and scripts required to reproduce the results of this work are publicly available on GitHub (github.com/RicardoReySaez/uSDT_pub, Rey-Sáez, 2026) and Zenodo (doi.org/10.5281/zenodo.24019945, Rey-Sáez et al., 2026).

Correspondence concerning this article should be addressed to Ricardo Rey-Sáez, Email: ricardoreysaez95@gmail.com, and Francisco Garre-Frutos, Email: francisco.garre@uam.es

SM1: Performance Measures in the Simulation Study

For each condition, let θ be the true parameter value and $\hat{\theta}_r$ and $\widehat{\mathrm{SE}}_r$ its estimate and SE in replication $r = 1, \ldots, R_u$, where R_u is the number of replications in which the parameter was evaluated. We computed bias and root mean squared error (RMSE),

$$\text{Bias} = \frac{1}{R_u}\sum_{r=1}^{R_u}\hat{\theta}_r - \theta, \qquad \text{RMSE} = \sqrt{\frac{1}{R_u}\sum_{r=1}^{R_u}(\hat{\theta}_r - \theta)^2}, \tag{S1}$$

and relative bias (bias divided by θ) for SDs, whose true values are always positive. The empirical coverage rate (ECR) was the proportion of 95% CIs containing θ, among replications with a CI. We assessed SEs using the mean squared relative SE (MSRSE, Chalmers & Adkins, 2020), the mean squared estimated SE divided by the variance of estimates across replications. A value of 1 indicates that SEs reflect sampling variability, whereas values above 1 indicate overestimation. For each hypothesis, the rejection rate among testable replications ($p < .05$) estimated Type I error when $\theta = 0$ and power otherwise. We also computed the rate across all replications, treating untestable fits as non-rejections, to estimate the probability of concluding in favor of the hypothesis.

A fit was untestable when `{uSDT}` could not compute SEs for the correlation and intercept due to a singular estimated random-effects covariance matrix. Means, the group-level difference, SDs, and covariance were evaluated whenever estimated, as was the correlation, including estimates at ±1. The intercept and slope were evaluated only in testable fits because their estimates become extremely unstable when σ_A approaches zero. Some estimates remained extreme even in testable fits, so we also report median bias (median estimate minus the true value).

To summarize agreement across conditions, we computed Lin's concordance correlation coefficient (CCC, Lin, 1989) between true values (x) and mean estimates (y),

$$\text{CCC} = \frac{2\,s_{xy}}{s_x^2 + s_y^2 + (\bar{x} - \bar{y})^2} = r \cdot C_b, \qquad C_b = \frac{2}{v + 1/v + u^2}, \tag{S2}$$

where s_x^2, s_y^2, and s_{xy} are variances and covariance computed using the number of conditions as divisor, r is the Pearson correlation, $v = s_x/s_y$ is a scale shift, and $u = (\bar{x} - \bar{y})/\sqrt{s_x s_y}$ is a location shift. The CCC combines precision (r) and accuracy (C_b), reaching 1 only when all points lie on the identity line. Because it depends on the range of true values, we use it only to summarize figures.

With 1,000 replications, the Monte Carlo SE of a proportion p is $\sqrt{p(1-p)/1{,}000}$, approximately .007 at $p = .05$ or $p = .95$. Rates based on fewer testable replications have larger Monte Carlo errors. We assessed coverage and Type I error against Bradley's liberal criterion (Bradley, 1978), [.925, .975] and [.025, .075], respectively.

To assess the influence of design factors on each measure, we fitted ANOVA meta-models (Skrondal, 2000) with the measure as the dependent variable and design factors as predictors. In Study 1, models included all five factors and interactions up to fourth order, leaving the fifth-order interaction as residual. In Study 2, models were fitted separately for each generating model, with interactions up to third order because the 324 conditions per model provided few residual degrees of freedom for higher-order terms. We report the non-partial omega-squared,

$$\omega^2 = \frac{SS_{\text{effect}} - df_{\text{effect}} \cdot MS_{\text{residual}}}{SS_{\text{total}} + MS_{\text{residual}}}, \tag{S3}$$

which quantifies the proportion of across-condition variation explained by each effect. Because each measure was averaged over 1,000 replications, residual variation mainly reflected Monte Carlo error, causing partial effect sizes (computed relative to this residual) to approach 1 for most effects. We interpret ω^2 values of .01, .06, and .14 as small, medium, and large (Cohen, 1992), and report effects with $\omega^2 \geq .06$ in at least one measure. However, a large ω^2 can be practically negligible when the measure varies little across conditions. Meta-models should therefore be interpreted alongside marginal means.

SM2: Simulation Study 1 (Validation)

True Values

The true SDs depend on reliability (R) and the number of trials per condition (K). For a participant with sensitivity d and criterion c, the expected hit and false-alarm rates are $\text{HR} = \Phi(\eta_S)$ and $\text{FAR} = \Phi(\eta_N)$, where $\eta_S = d'/2 - c$ and $\eta_N = -d'/2 - c$. Let $Z_{\text{HR}} = \Phi^{-1}(\widehat{\text{HR}})$ and $Z_{\text{FAR}} = \Phi^{-1}(\widehat{\text{FAR}})$ be the probit-transformed observed rates. With K_S signal and K_N noise trials, their sampling variances are approximately

$$\mathbb{V}(Z_{\text{HR}}) \approx \frac{\text{HR} \cdot (1 - \text{HR})}{K_S \cdot \phi(\eta_S)^2} \qquad \text{and} \qquad \mathbb{V}(Z_{\text{FAR}}) \approx \frac{\text{FAR} \cdot (1 - \text{FAR})}{K_N \cdot \phi(\eta_N)^2}, \tag{S4}$$

where ϕ is the standard normal density function (Gourevitch & Galanter, 1967, p. 31; see also Suero et al., 2017). When the criterion is estimated, as in the awareness task, sensitivity is estimated as the difference between the two transformed rates. Because these are independent, its sampling variance is the sum of their variances,

$$\hat{d} = Z_{\text{HR}} - Z_{\text{FAR}}, \qquad \mathbb{V}(\hat{d}) \approx \mathbb{V}(Z_{\text{HR}}) + \mathbb{V}(Z_{\text{FAR}}).$$

When the criterion is fixed at zero, as in the influence task after dichotomization, $\eta_S = d/2$ and $\eta_N = -d/2$. Each rate then provides an independent estimate of sensitivity, $\hat{d}_S = 2 \cdot Z_{\text{HR}}$ and $\hat{d}_N = -2 \cdot Z_{\text{FAR}}$, with sampling variances of $\mathbb{V}(\hat{d}_S) \approx 4 \cdot \mathbb{V}(Z_{\text{HR}})$ and $\mathbb{V}(\hat{d}_N) \approx 4 \cdot \mathbb{V}(Z_{\text{FAR}})$. Asymptotically, the maximum likelihood estimate of d and its sampling variance are

$$\hat{d} \approx \frac{\hat{d}_S/\mathbb{V}(\hat{d}_S) + \hat{d}_N/\mathbb{V}(\hat{d}_N)}{1/\mathbb{V}(\hat{d}_S) + 1/\mathbb{V}(\hat{d}_N)} \qquad \text{and} \qquad \mathbb{V}(\hat{d}) \approx \frac{4}{1/\mathbb{V}(Z_{\text{HR}}) + 1/\mathbb{V}(Z_{\text{FAR}})}.$$

The reliability of the observed sensitivities is then

$$R = \frac{\sigma^2}{\sigma^2 + \mathbb{E}[\mathbb{V}(\hat{d})]}, \tag{S5}$$

where the expectation is over the population distribution of true sensitivities and criteria. For each condition, we numerically solved Equation S5 for σ_A and σ_I. Because sampling variance depends on sensitivity, and μ_I depends on σ_I through the intercept (Equation 2 in the main text), we iterated σ_I and μ_I until convergence. Table S1 reports the resulting SDs. For a given reliability, more trials imply smaller true SDs because measurement error decreases.

Given σ_A, σ_I, ρ, and β_0, we computed the slope and influence-task mean as

$$\beta_1 = \rho \cdot \frac{\sigma_I}{\sigma_A}, \qquad \mu_I = \beta_0 + \beta_1 \cdot \mu_A.$$

The residual SD of true influence-task sensitivity was $\sigma_I \cdot \sqrt{1 - \rho^2}$.

Table S1

True Standard Deviations of Sensitivities in Simulation Study 1

	σ_A			σ_I		
R	$K = 10$	$K = 30$	$K = 50$	$K = 10$	$K = 30$	$K = 50$
.1	0.19	0.11	0.09	0.19	0.11	0.08
.2	0.29	0.16	0.13	0.28	0.16	0.13
.3	0.38	0.22	0.17	0.37	0.21	0.16
.4	0.47	0.27	0.21	0.46	0.27	0.21
.5	0.58	0.33	0.26	0.57	0.33	0.25
.6	0.72	0.41	0.31	0.71	0.40	0.31
.7	0.91	0.51	0.39	0.89	0.50	0.39

Note. R = reliability of the observed sensitivities; K = trials per condition in each task. σ_I is averaged over ρ and β_0, which change it by less than .02.

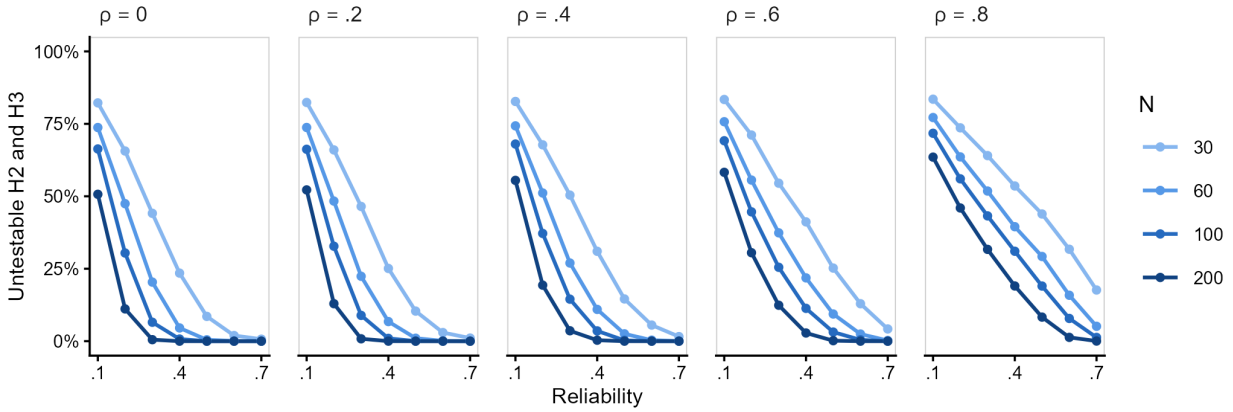
Convergence and Boundary Fits

Of the 1,260,000 fits, only 106 failed with an estimation error. Across conditions, an average of 23% of fits were singular (maximum 80%), and 25% were untestable for H2 and H3 (maximum 86%). All singular fits were untestable, but another 2% were untestable despite being nonsingular. The most common boundary estimates were correlations of ±1 (average 22%; maximum 70%), followed by $\sigma_A = 0$ (average 3%; maximum 23%) and $\sigma_I = 0$ (average 1%; maximum 14%). The SD of the awareness-task criterion was almost never estimated at zero (0.1%).

Figure S1 shows how the proportion of untestable fits varied with reliability, sample size, and true correlation. The ANOVA meta-models showed that reliability explained most of the variation across conditions ($\omega^2 = .75$), followed by sample size ($\omega^2 = .12$) and true correlation ($\omega^2 = .06$; Table S3). Untestable fits were more frequent with stronger correlations because their estimates were more likely to reach ±1. At $\rho = .8$, they remained frequent even with moderate reliability.

Figure S1

Untestable Fits by Reliability, Sample Size, and True Correlation in Simulation Study 1



Note. Proportion of fits without SEs for the correlation and the intercept, averaged over trials per condition and true intercepts.

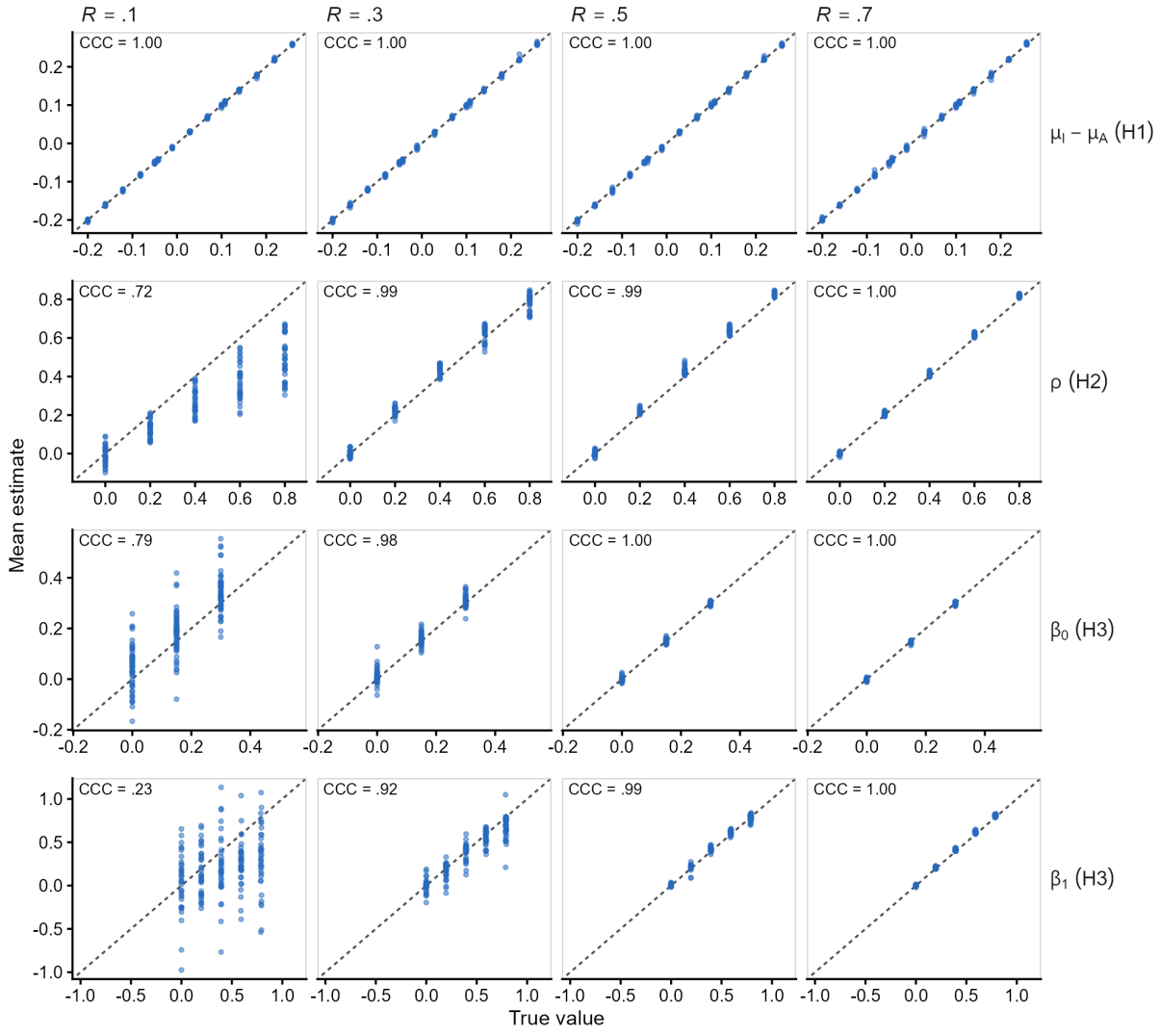
Recovery of All Parameters

Table S2 summarizes parameter recovery. The means of both tasks, the mean awareness-task criterion, and the group-level difference showed negligible bias (absolute bias below .017 in every condition), coverage between .905 and .970, and MSRSE between 0.84 and 1.19. The SDs of true sensitivities were slightly underestimated, as is typical of maximum likelihood estimates of variance components. At low reliability, relative bias reached $-$22% for σ_A and $-$13% for σ_I (see the section on random-effect SDs).

The intercept and slope showed the largest mean biases, ranging from $-$0.23 to 0.27 for β_0 and from $-$1.32 to 0.74 for β_1. However, their median biases were much smaller, ranging from $-$0.02 to 0.15 and from $-$0.71 to 0.09, respectively. Thus, most mean bias came from a few extreme estimates at low reliability, when σ_A, which appears in the denominator of the slope, was estimated near zero. A few extreme SEs also inflated the MSRSE of the correlation, intercept, and slope in these conditions, making coverage a better indicator of CI calibration. Figure S2 shows the recovery of the parameters of all three hypotheses, including the slope, at four reliability levels.

Figure S2

Recovery of the Parameters of the Three Hypotheses by Reliability in Simulation Study 1



Note. Each point is one condition, showing the mean estimate across 1,000 replications against the true value, for four of the seven reliability levels (*R*). Dashed lines indicate perfect recovery. CCC = Lin's concordance correlation coefficient within each panel. The intercept and the slope were evaluated over testable fits.

Table S2

Recovery of All Parameters in Simulation Study 1

Parameter	Bias	RMSE	ECR	MSRSE
μ_A	0.001 [−0.010, 0.014]	[0.019, 0.213]	.944 [.912, .965]	0.98 [0.95, 1.02]
μ_I	0.000 [−0.009, 0.017]	[0.018, 0.203]	.946 [.914, .970]	0.99 [0.96, 1.03]
μ_{cA}	0.000 [−0.007, 0.007]	[0.022, 0.082]	.942 [.905, .966]	0.97 [0.94, 1.01]
σ_A	−0.017 [−0.061, 0.006]	[0.025, 0.196]	.950 [.767, .994]	1.12 [0.99, 1.93]
σ_I	−0.011 [−0.048, 0.013]	[0.024, 0.187]	.946 [.740, .990]	1.13 [1.00, 2.01]
σ_{cA}	−0.008 [−0.033, 0.000]	[0.017, 0.098]	.957 [.898, .995]	0.98 [0.94, 1.02]
σ_{AI}	−0.002 [−0.024, 0.012]	[0.004, 0.259]	.962 [.891, 1.000]	1.09 [1.01, 1.43]
$\mu_I - \mu_A$	0.000 [−0.014, 0.014]	[0.025, 0.288]	.946 [.917, .970]	0.99 [0.96, 1.03]
ρ	−0.012 [−0.495, 0.090]	[0.061, 1.023]	.973 [.848, 1.000]	1.98 [1.02, 8.89]
β_0	0.009 [−0.229, 0.269]	[0.033, 3.687]	.960 [.889, 1.000]	3.22 [1.04, 324.35]
β_1	−0.043 [−1.325, 0.740]	[0.093, 13.204]	.962 [.867, 1.000]	4.97 [1.08, 380.11]

Note. Mean [minimum, maximum] over the 1,260 conditions for bias and ECR, [minimum, maximum] for RMSE, and median [first quartile, third quartile] for MSRSE, which took extreme values in a few conditions with very low reliability. ECR = empirical coverage rate of the 95% CIs; MSRSE = mean squared relative SE. μ_{cA} and σ_{cA} are the mean and SD of the awareness-task criterion. The intercept and the slope were evaluated over testable fits.

Meta-Models

Table S3 and Table S4 report the ANOVA meta-model effect sizes, while Table S5 and Table S6 report the corresponding marginal means.

Reliability was the main determinant of bias and coverage. For the SDs of true sensitivities, relative bias depended on reliability (ω^2 = .45 for σ_A and .30 for σ_I) and its interaction with sample size (ω^2 = .29 and .30). Larger samples reduced bias, except at very low reliability, when many SD estimates were zero. Coverage depended almost entirely on reliability (ω^2 = .73 and .81), whereas RMSE depended on trials (ω^2 = .56 and .54) and sample size (ω^2 = .36 and .38). Thus, at fixed reliability, trials affected precision but not bias or coverage.

The group-level difference showed no effects on bias (all $\omega^2 < .02$), indicating unbiased estimates across the design. Its RMSE depended on trials and sample size (ω^2 = .43 and .39). Reliability had a large effect on coverage (ω^2 = .16), but this was practically negligible, as marginal coverage ranged only from .942 to .953 (Table S6).

For the correlation, bias depended on reliability (ω^2 = .48) and its interaction with the true correlation (ω^2 = .23). RMSE depended on reliability (ω^2 = .77) and sample size (ω^2 = .16). Coverage depended on the true correlation (ω^2 = .40) and its interaction with reliability (ω^2 = .35). At low reliability, CIs were conservative for weak correlations and liberal for strong correlations.

The intercept showed a similar pattern, with smaller effects on bias (reliability: ω^2 = .15; reliability by correlation: ω^2 = .12) and similar effects on coverage (reliability by correlation: ω^2 = .37; reliability: ω^2 = .25; correlation: ω^2 = .16).

Table S3

Effect Sizes of the Meta-Models for the Random-Effect Standard Deviations and Boundary Fits in Simulation Study 1

	σ_A			σ_I			Boundary
Predictor	RB	RMSE	ECR	RB	RMSE	ECR	UNT
N	0.05	**0.36**	0.01	**0.14**	**0.38**	0.01	0.12
K	0.04	**0.56**	0.00	0.00	**0.54**	0.00	0.00
R	**0.45**	0.01	**0.73**	**0.30**	0.00	**0.81**	**0.75**
ρ	0.04	0.00	0.00	0.08	0.00	0.00	0.06
$N \times R$	**0.29**	0.02	**0.16**	**0.30**	0.01	0.10	0.04

Note. Non-partial ω^2. N = sample size; K = trials per condition; R = reliability; ρ = true correlation; β_0 = true intercept. RB = relative bias; RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs; UNT = untestable fits for H2 and H3. Effects with $\omega^2 < .06$ in every measure are omitted, and values of at least .14 are in bold.

Table S4

Effect Sizes of the Meta-Models for the Three Hypotheses in Simulation Study 1

	H1: $\mu_I - \mu_A$			H2: ρ			H3: β_0		
Predictor	Bias	RMSE	ECR	Bias	RMSE	ECR	Bias	RMSE	ECR
N	0.00	**0.39**	0.02	0.03	**0.16**	0.06	0.03	0.04	0.01
K	0.00	**0.43**	0.00	0.00	0.00	0.00	0.00	0.03	0.00
R	0.00	0.07	**0.16**	**0.48**	**0.77**	0.04	**0.15**	**0.36**	**0.25**
ρ	0.01	0.02	0.04	0.05	0.03	**0.40**	0.06	0.00	**0.16**
$N \times R$	0.01	0.01	0.03	0.09	0.02	0.00	0.03	0.01	0.01
$R \times \rho$	0.00	0.01	0.01	**0.23**	0.01	**0.35**	0.12	0.00	**0.37**
$N \times R \times \rho$	0.00	0.00	0.00	0.04	0.01	0.07	0.00	0.00	0.03

Note. Non-partial ω^2. N = sample size; K = trials per condition; R = reliability; ρ = true correlation; β_0 = true intercept. RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs. Effects with $\omega^2 < .06$ in every measure are omitted, and values of at least .14 are in bold.

Table S5

Marginal Means for the Random-Effect Standard Deviations and Boundary Fits in Simulation Study 1

	σ_A			σ_I			Boundary
Level	RB	RMSE	ECR	RB	RMSE	ECR	UNT
Sample size (N)							
30	−0.072	0.112	0.944	−0.054	0.107	0.941	0.393
60	−0.060	0.083	0.948	−0.038	0.079	0.945	0.272
100	−0.055	0.066	0.952	−0.032	0.063	0.948	0.206
200	−0.047	0.049	0.955	−0.023	0.045	0.950	0.137
Trials per condition (K)							
10	−0.070	0.117	0.953	−0.039	0.111	0.947	0.264
30	−0.052	0.064	0.949	−0.035	0.062	0.945	0.248
50	−0.054	0.051	0.948	−0.036	0.048	0.946	0.244
Reliability (R)							
0.1	−0.081	0.079	**0.876**	−0.015	0.075	**0.860**	0.705
0.2	**−0.101**	0.082	0.940	−0.061	0.075	0.934	0.466
0.3	−0.078	0.079	0.964	−0.055	0.073	0.959	0.283
0.4	−0.056	0.075	0.972	−0.045	0.071	0.971	0.164
0.5	−0.041	0.073	0.973	−0.034	0.071	0.971	0.088
0.6	−0.031	0.074	0.965	−0.026	0.072	0.966	0.042
0.7	−0.022	0.079	0.959	−0.020	0.077	0.959	0.016
True correlation (ρ)							
0	−0.066	0.079	0.952	−0.045	0.075	0.948	0.193
0.2	−0.065	0.079	0.951	−0.043	0.075	0.947	0.201
0.4	−0.061	0.078	0.951	−0.040	0.074	0.947	0.222
0.6	−0.056	0.077	0.950	−0.033	0.073	0.945	0.269
0.8	−0.044	0.075	0.946	−0.022	0.071	0.942	0.375
True intercept (β_0)							
0	−0.059	0.077	0.950	−0.037	0.073	0.946	0.252
0.15	−0.058	0.077	0.950	−0.037	0.073	0.946	0.252
0.3	−0.059	0.077	0.950	−0.037	0.074	0.945	0.252

Note. Means over the remaining factors. RB = relative bias; RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs; UNT = untestable fits for H2 and H3. Relative biases of at least 10% and coverages outside [.925, .975] are in bold.

Table S6

Marginal Means for the Three Hypotheses in Simulation Study 1

	H1: $\mu_I - \mu_A$			H2: ρ			H3: β_0		
Level	Bias	RMSE	ECR	Bias	RMSE	ECR	Bias	RMSE	ECR
Sample size (N)									
30	−0.000	0.123	0.944	−0.034	0.551	**0.982**	0.020	0.427	0.963
60	−0.001	0.087	0.946	−0.015	0.431	0.974	0.006	0.338	0.961
100	−0.000	0.067	0.947	−0.005	0.355	0.970	0.006	0.265	0.959
200	−0.001	0.047	0.947	0.006	0.267	0.965	0.003	0.185	0.958
Trials per condition (K)									
10	−0.001	0.121	0.946	−0.012	0.409	0.974	0.006	0.407	0.960
30	−0.000	0.069	0.946	−0.011	0.399	0.973	0.009	0.257	0.960
50	−0.000	0.053	0.947	−0.013	0.395	0.971	0.011	0.246	0.961
Reliability (R)									
0.1	−0.001	0.068	0.953	−0.148	0.832	**0.980**	0.040	0.867	0.971
0.2	−0.001	0.070	0.949	−0.027	0.610	0.974	0.021	0.505	0.970
0.3	−0.001	0.074	0.946	0.016	0.441	**0.975**	0.007	0.300	0.968
0.4	−0.000	0.078	0.946	0.026	0.327	0.975	0.001	0.158	0.963
0.5	−0.001	0.083	0.944	0.024	0.247	0.973	−0.002	0.112	0.957
0.6	−0.000	0.091	0.943	0.016	0.194	0.968	−0.003	0.090	0.949
0.7	−0.000	0.103	0.942	0.010	0.156	0.962	−0.002	0.092	0.945
True correlation (ρ)									
0	−0.001	0.090	0.944	−0.000	0.448	**0.983**	−0.000	0.339	0.968
0.2	−0.001	0.085	0.946	0.002	0.442	**0.983**	0.001	0.310	0.966
0.4	−0.001	0.081	0.945	0.002	0.416	**0.982**	0.005	0.318	0.963
0.6	−0.000	0.077	0.946	−0.016	0.378	**0.975**	0.013	0.281	0.957
0.8	−0.000	0.072	0.949	−0.048	0.320	0.939	0.025	0.269	0.947
True intercept (β_0)									
0	−0.001	0.081	0.946	−0.013	0.401	0.973	0.010	0.324	0.960
0.15	−0.001	0.081	0.946	−0.011	0.401	0.973	0.008	0.289	0.960
0.3	−0.000	0.081	0.946	−0.012	0.401	0.972	0.009	0.297	0.960

Note. Means over the remaining factors. RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs. Coverages outside [.925, .975] are in bold.

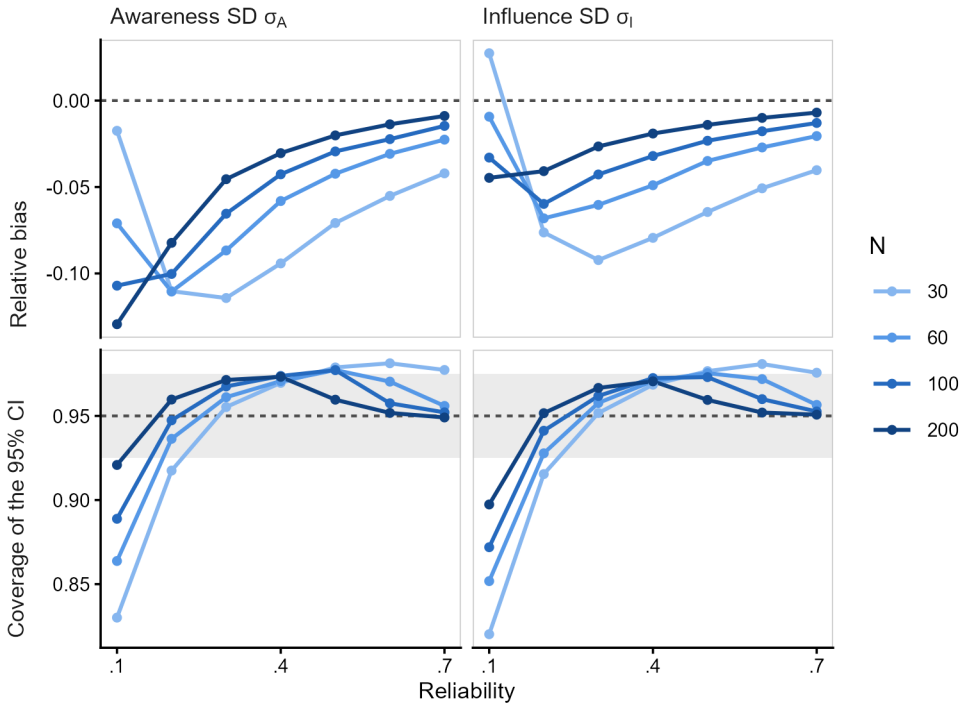
Random-Effect Standard Deviations

Figure S3 shows the relative bias and coverage of the SDs by reliability and sample size. Both SDs were underestimated, by less than 5% when reliability was at least .4 and $N \geq 100$, but by up to 13% on average at low reliability.

At reliability of .1, many estimates were at or near zero, reducing mean coverage to .87, because Wald CIs on the log scale cannot cover the true value when estimates approach zero. For the same reason, SEs overestimated sampling variability (median MSRSE = 3.29). At reliability of .4 or higher, MSRSE was close to 1 (median values from 0.98 to 1.11).

Figure S3

Relative Bias and Coverage of the Random-Effect Standard Deviations in Simulation Study 1



Note. Means over trials per condition, true correlations, and true intercepts. The gray band in the coverage panels shows Bradley's liberal criterion, [.925, .975].

Correlation

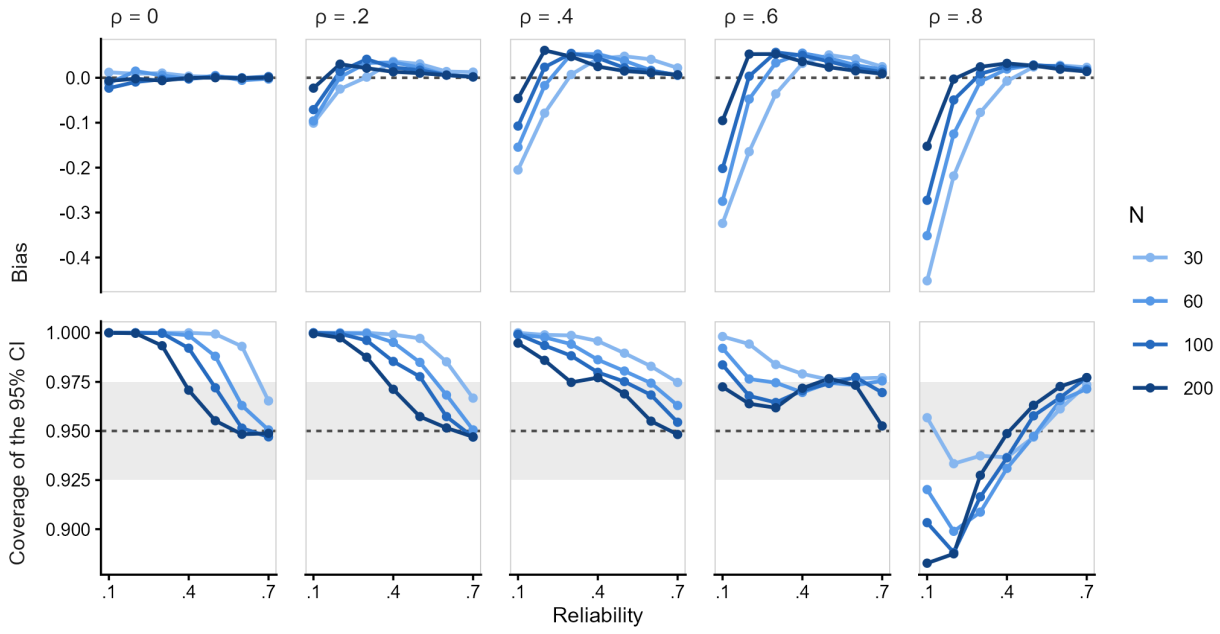
Figure S4 shows correlation bias and coverage by true correlation, reliability, and sample size. Estimates were unbiased when the true correlation was zero. At low reliability, strong correlations were underestimated because estimates were highly variable but bounded at ±1. At

moderate reliability, they were slightly overestimated (by .02 to .06), with bias disappearing as reliability increased. At low reliability, 95% CIs were conservative for weak correlations, with coverage close to 1. Coverage fell below .925 only at $\rho = .8$ and reliability of .3 or lower.

Because CIs and tests of H2 were available only for testable fits, inferences were based on a subset of replications selected according to their estimates. Table S7 compares correlation bias across all fits and testable fits. At very low reliability, boundary fits often had correlations of ±1, so excluding them increased the underestimation of strong correlations. With $R = .1$ and $\rho = .8$, bias changed from −.31 to −.54. At reliability of .5 or higher, both biases were close to zero.

Figure S4

Bias and Coverage of the Correlation by True Correlation in Simulation Study 1



Note. Means over trials per condition and true intercepts. Bias was computed over all fits with an estimate, and coverage over testable fits. The gray band in the coverage panels shows Bradley's liberal criterion, [.925, .975].

Hypothesis Tests

Figure S5 shows H1 rejection rates against the true group-level difference. Although no condition had a difference of exactly zero, in the 84 conditions closest to zero (−.011), rejection rates ranged from .037 to .086, close to the nominal .05.

Table S7

Bias of the Correlation Over All Fits and Over Testable Fits in Simulation Study 1

	ρ = .2		ρ = .4		ρ = .6		ρ = .8	
R	All	Testable	All	Testable	All	Testable	All	Testable
.1	−.07	−.14	−.13	−.27	−.22	−.40	−.31	−.54
.2	.01	−.07	.00	−.13	−.04	−.22	−.10	−.33
.3	.02	−.02	.04	−.05	.03	−.10	−.01	−.18
.4	.03	.00	.04	−.01	.04	−.04	.02	−.09
.5	.02	.01	.03	.01	.04	.00	.03	−.04
.6	.01	.01	.02	.01	.03	.01	.02	−.01
.7	.00	.00	.01	.01	.02	.01	.02	.01

Note. Means over sample sizes, trials per condition, and true intercepts. All = every fit with an estimate, including estimates at ±1; Testable = fits with SEs for the correlation. The correlation was unbiased under both definitions when $\rho = 0$.

Figure S6 shows H1 power for the 15 true differences by sample size and number of trials, averaged over reliability. Power increased with the size of the difference, sample size, and trials. For example, with a difference of −.20 and $N = 100$, power increased from 49% with 10 trials to 97% with 50 trials. Because each difference corresponds to a combination of intercept and correlation, power also increased with correlation, which reduces the variance of the difference between sensitivities. Thus, a difference of .108 ($\rho = .8$) was detected more often than one of .100 ($\rho = 0$).

The figure shows trials rather than reliability because they affected H1 differently in our design. At a fixed number of trials, the sampling variance of d' is fixed, so higher reliability requires larger true SDs (Table S1). Consequently, the variance of observed sensitivities (true plus sampling variance) increased with reliability. With 10 trials, the awareness-task variance increased from 0.36 at $R = .1$ to 1.19 at $R = .7$. Larger individual differences increase the variability of the group-level mean. Thus, at fixed trials, H1 power was lower with $R = .7$ than with $R = .1$ in 94% of conditions with a difference of at least .05. Across reliability levels, power varied by .14 on average (maximum .45).

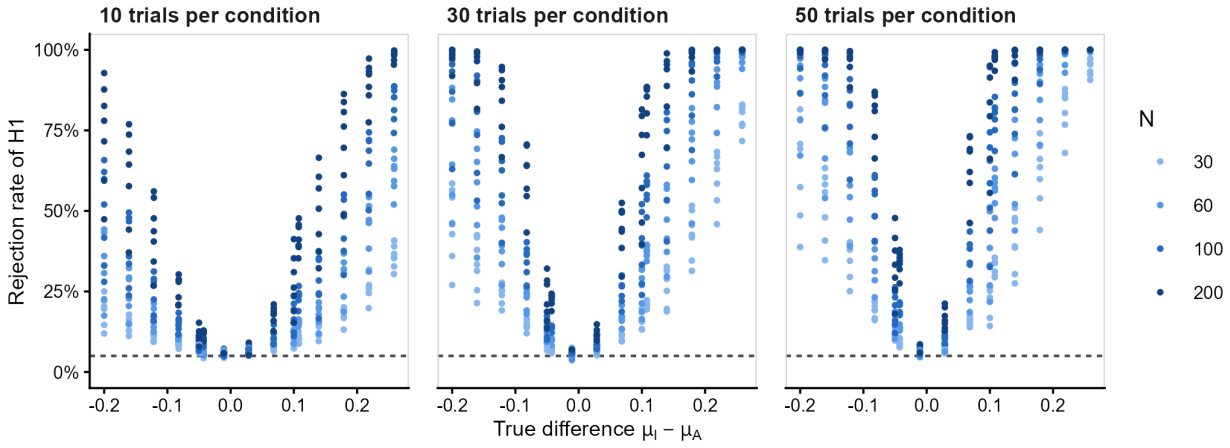
In contrast, more trials reduce sampling variance without changing the true SDs. Power was at least as high with 50 as with 10 trials in all of these conditions. Thus, H1 power depends on

the variance of the observed difference, not reliability itself. For H2 and H3, which concern individual differences, power increased with reliability (Figure 2 of the main text).

Figure S7 compares H2 and H3 rejection rates over testable fits and all replications. The Type I error rate of H2 was close to .05 with reliability of at least .4 and $N \geq 100$, but conservative otherwise, falling below .01 with $N = 30$ and reliability of .4 or lower. For H3, it was close to .05 except in the nine conditions with reliability of .2 or lower and $\rho = .8$ described in the main text. Power increased with sample size, reliability, and effect size. At low reliability, the gap between the two lines in each panel shows that most power loss came from untestable fits. At $R = .1$, researchers would rarely be able to test H2 or H3, regardless of sample size.

Figure S5

Rejection Rate of H1 by True Difference in Simulation Study 1

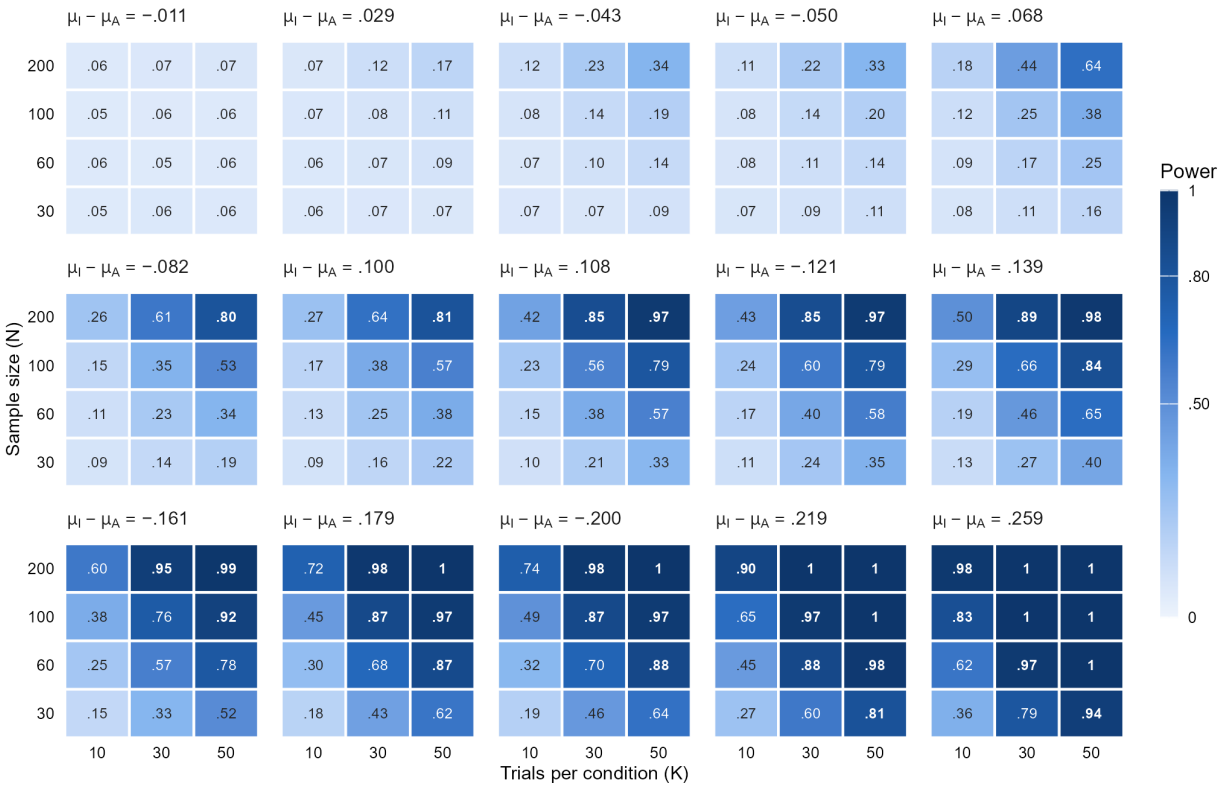


Note. Each point is one condition. The dashed line shows the nominal .05 level. H1 was always testable.

Figure 2 of the main text shows power counting untestable fits as non-rejections, which is the probability that a researcher concludes in favor of the hypothesis when planning a study under each condition. Figure S8 shows power over testable fits only. For correlations up to .6 with $R \geq .4$, both versions differed by at most .08. Power over testable fits was much higher where boundary fits were frequent: for $\rho = .8$, for which many estimates lay at +1 (up to .28 higher), and for H3 with $R \leq .3$ (up to .36 higher). These values are conditional on obtaining a testable fit, which a researcher cannot guarantee, and with low reliability testable fits are a selected subset, the

Figure S6

Power of H1 by Trials, Sample Size, and True Difference in Simulation Study 1

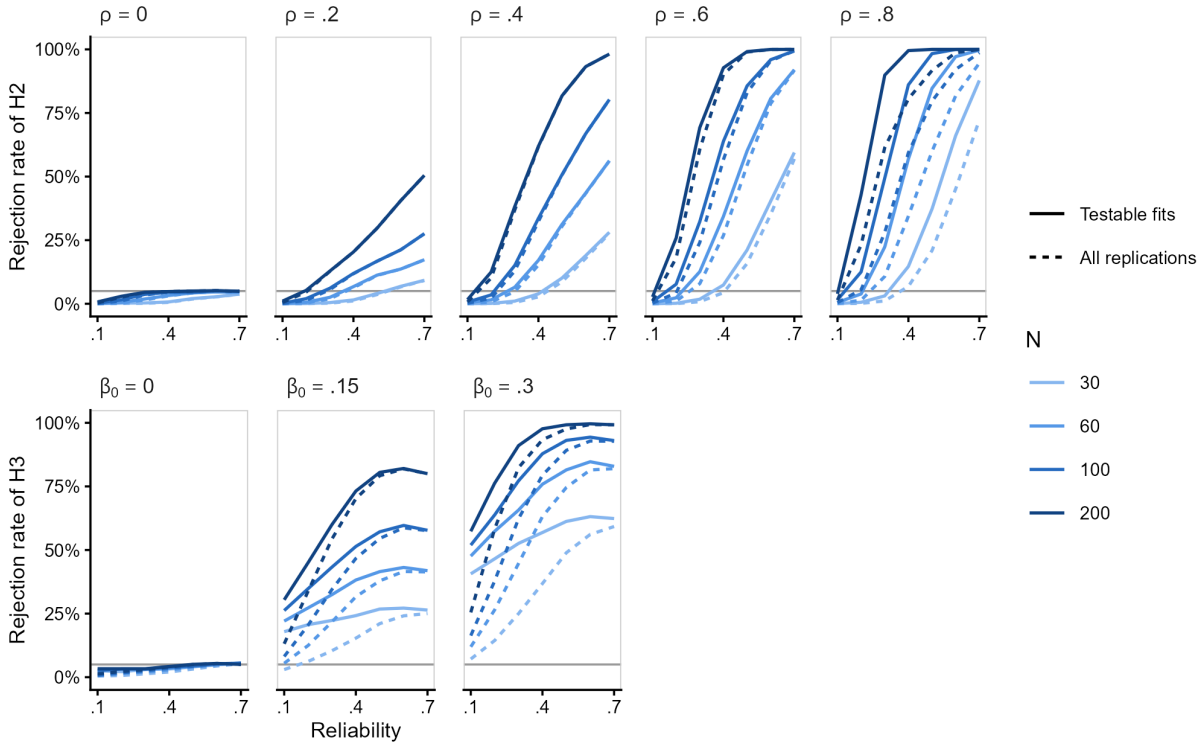


Note. Proportion of the 1,000 replications per condition in which H1 was rejected at $\alpha = .05$, averaged over the seven reliability levels, with which power decreased slightly for a fixed number of trials (see text). Each panel shows one of the 15 true group-level differences, ordered by absolute value; the first one, closest to zero, shows a rate close to the Type I error rate. Values of .80 or more are in bold. H1 was always testable.

same in which the Type I error rate of H3 exceeded .075 in some conditions. We therefore report power over all replications in the main text.

Figure S7

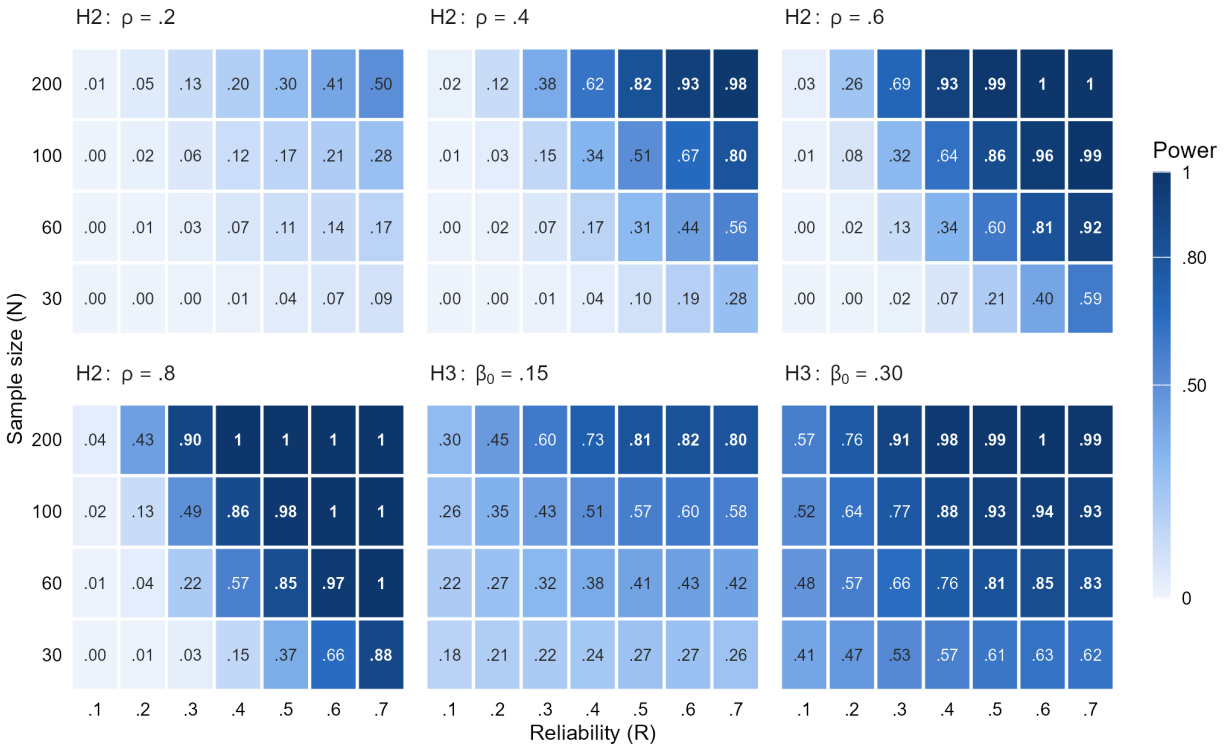
Rejection Rates of H2 and H3 in Simulation Study 1



Note. Means over trials per condition and the true value of the other parameter. Solid lines show rejection rates over testable fits, and dashed lines over all replications, counting untestable fits as non-rejections. Panels with $\rho = 0$ and $\beta_0 = 0$ show Type I error rates; the gray line marks .05.

Figure S8

Power Over Testable Fits by Reliability, Sample Size, and Effect Size in Simulation Study 1



Note. Proportion of testable fits in which H2 or H3 was rejected at $\alpha = .05$. Each H2 panel shows one nonzero true correlation, averaged over the three true intercepts, and each H3 panel one nonzero true intercept, averaged over the five true correlations; all cells are also averaged over trials per condition. Values of .80 or more are in bold. Figure 2 of the main text shows the same power counting untestable fits as non-rejections.

SM3: Simulation Study 2 (Robustness)

Data-Generating Models and Population Values

This simulation examined the robustness of the multilevel SDT model when the true sensitivities do not follow a bivariate normal distribution. We generated trial-level data under two alternative population models, adapted from Klauer et al. (1998) and Goldstein et al. (2022). Both models allow a proportion of participants to have zero awareness sensitivity, but differ in how awareness sensitivities are distributed and how they relate to influence-task sensitivity.

Klauer et al. (1998). True awareness-task sensitivity follows a normal distribution censored at zero:

$$d'_{iA} = \max(0, M_i), \qquad M_i \sim \text{Normal}(\mu = m, \sigma = s). \tag{S6}$$

Thus, participants whose latent values are negative have zero awareness sensitivity, while the remaining participants have positive sensitivities. We fixed $s = 2$ and chose m to obtain the desired population proportion of unaware participants, $\pi_0 = \Phi(-m/s)$. True influence-task sensitivity was generated as

$$d'_{iI} = \alpha + \beta \cdot d'_{iA} + \varepsilon_i, \tag{S7}$$

where ε_i is independent normally distributed residual variation. This residual was added to Klauer et al.'s (1998) original specification to allow correlations between true sensitivities smaller than one. Because the relationship is linear in d'_{iA}, α represents the expected influence-task sensitivity when awareness sensitivity is zero.

Goldstein et al. (2022). True awareness-task accuracy, θ_i, follows a mixture distribution. A proportion π_0 of participants has chance-level accuracy ($\theta_i = .5$), while the remaining participants have accuracies drawn uniformly between .5 and 1. Assuming an unbiased observer in a standard SDT model, these accuracies correspond to sensitivities

$$d'_{iA} = 2 \cdot \Phi^{-1}(\theta_i). \tag{S8}$$

In contrast to the Klauer model, true influence-task sensitivity was generated as a linear function of awareness accuracy above chance (θ_i), not sensitivity (d'_i):

$$d'_{iI} = \alpha + \beta \cdot (\theta_i - .5) + \varepsilon_i. \tag{S9}$$

Consequently, α again represents the expected influence-task sensitivity of unaware participants, but the relationship is not linear when awareness is expressed as d'_{iA}.

Trial-level data. In both models, awareness-task responses were generated using a standard probit SDT model, with individual response criteria drawn from $\mathcal{N}(0, .3^2)$. For the Klauer model, influence-task responses were also binary. For the Goldstein model, we generated normally distributed RTs with unit within-condition SD and a standardized difference between conditions of d'_{iI}. These RTs were dichotomized at each participant's median, following Meyen et al. (2022), before fitting the multilevel SDT model. In both models, we fixed the population SD of influence-task sensitivity at $\sigma_I = .3$ and selected β and the residual SD to obtain the desired correlation ρ between influence-task sensitivity and its generating predictor.

Population values. The multilevel SDT model describes the joint distribution of true sensitivities using their means, SDs, and correlation. We calculated these population values under each generating model to evaluate parameter recovery, focusing particularly on the correlation and regression intercept. The latter is defined as

$$\beta_0 = \mu_I - \frac{\mathrm{Cov}(d'_{iA}, d'_{iI})}{\mathbb{V}(d'_{iA})} \cdot \mu_A, \tag{S10}$$

which defines the intercept of the best linear prediction of d'_{iI} from d'_{iA}.

Under the Klauer model, the relationship between true sensitivities is linear, so the population regression intercept equals the generating intercept, $\beta_0 = \alpha$, and the correlation between true sensitivities equals ρ.

Under the Goldstein model, however, influence-task sensitivity is linear in awareness accuracy rather than in d' (Equation S9). Recall that awareness can be expressed either as

accuracy above chance, $\theta_i - .5$, or as SDT sensitivity. Under the equal-variance SDT model with an unbiased response criterion, accuracy is given by $\theta_i = \Phi(d'_{iA}/2)$, so solving for d'_{iA} gives $d'_{iA} = 2 \cdot \Phi^{-1}(\theta_i)$ (Equation S8). Although these two measures represent awareness on different scales, their relationship is nonlinear. To determine the corresponding population intercept and correlation on the d' scale, we need to calculate their means, variances, and covariance.

For simplicity, let $X = d'_{iA}$ and $U = \theta_i - .5$. Under the Goldstein model, their expected values are

$$\begin{aligned} \mathbb{E}(X) &= p \cdot \mathbb{E}(X \mid \text{aware}) + (1 - p) \cdot \mathbb{E}(X \mid \text{unaware}), \\ \mathbb{E}(U) &= p \cdot \mathbb{E}(U \mid \text{aware}) + (1 - p) \cdot \mathbb{E}(U \mid \text{unaware}). \end{aligned} \tag{S11}$$

Notice, however, that both awareness sensitivity and accuracy above chance are exactly zero for unaware participants. Therefore, Equation S11 simplifies to

$$\mathbb{E}(X) = p \cdot \mathbb{E}(X \mid \text{aware}) \qquad \text{and} \qquad \mathbb{E}(U) = p \cdot \mathbb{E}(U \mid \text{aware}). \tag{S12}$$

For aware participants, U follows a uniform distribution between 0 and .5. Its expected value and variance are therefore

$$\begin{aligned} \mathbb{E}(U \mid \text{aware}) &= \frac{a + b}{2} = \frac{0 + 0.5}{2} = \frac{1}{4}, \\ \mathbb{V}(U \mid \text{aware}) &= \frac{(b - a)^2}{12} = \frac{(0.5 - 0)^2}{12} = \frac{1}{48}. \end{aligned} \tag{S13}$$

For aware participants, X follows a half-normal distribution with scale parameter 2, corresponding to a normal distribution with mean 0 and SD 2 truncated below at zero. The expected value and variance of a lower-truncated normal distribution are (Greene, 2011, p. 836):

$$\mathbb{E}(X \mid \text{aware}) = \mu_X + \sigma_X \cdot \lambda(\tau), \qquad \text{and} \qquad \mathbb{V}(X \mid \text{aware}) = \sigma_X^2 \cdot [1 - \delta(\tau)], \tag{S14}$$

where $\tau = -\mu_X/\sigma_X$,

$$\lambda(\tau) = \frac{\phi(\tau)}{1 - \Phi(\tau)}, \qquad \text{and} \qquad \delta(\tau) = \lambda(\tau) \cdot [\lambda(\tau) - \tau]\,. \tag{S15}$$

Here, ϕ and Φ are the probability density function and cumulative distribution function of the standard normal distribution, respectively. In our case, $\mu_X = 0$, $\sigma_X = 2$, and $\tau = 0$, so Equation S14 gives

$$\mathbb{E}(X \mid \text{aware}) = 2 \cdot \sqrt{\frac{2}{\pi}}, \qquad \text{and} \qquad \mathbb{V}(X \mid \text{aware}) = 4 \cdot \left(1 - \frac{2}{\pi}\right). \tag{S16}$$

To obtain the population means and variances, we account for the proportion p of aware participants. As shown in Equation S12, the population means are obtained by multiplying the conditional means (Equation S13 and Equation S16) by p:

$$\mathbb{E}(X) = p \cdot 2 \cdot \sqrt{\frac{2}{\pi}} \qquad \text{and} \qquad \mathbb{E}(U) = p \cdot \frac{1}{4}. \tag{S17}$$

The population variances can then be obtained using the identity $\mathbb{V}(Y) = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2$. Since both variables are zero for unaware participants, their second moments are

$$\mathbb{E}(X^2) = p \cdot \mathbb{E}(X^2 \mid \text{aware}), \qquad \text{and} \qquad \mathbb{E}(U^2) = p \cdot \mathbb{E}(U^2 \mid \text{aware}). \tag{S18}$$

Using $\mathbb{E}(Y^2 \mid \text{aware}) = \mathbb{V}(Y \mid \text{aware}) + [\mathbb{E}(Y \mid \text{aware})]^2$ and the conditional moments in Equation S13 and Equation S16, we obtain

$$\begin{aligned} \mathbb{E}(X^2 \mid \text{aware}) &= 4 \cdot \left(1 - \frac{2}{\pi}\right) + \left(2 \cdot \sqrt{\frac{2}{\pi}}\right)^2 = 4, \\ \mathbb{E}(U^2 \mid \text{aware}) &= \frac{1}{48} + \left(\frac{1}{4}\right)^2 = \frac{1}{12}. \end{aligned} \tag{S19}$$

Therefore, combining Equation S17, Equation S18, and Equation S19, the population

variances are

$$\begin{aligned} \mathbb{V}(X) &= 4 \cdot p - \left(2 \cdot p \cdot \sqrt{\frac{2}{\pi}}\right)^2 = 4 \cdot p \cdot \left(1 - \frac{2 \cdot p}{\pi}\right), \\ \mathbb{V}(U) &= \frac{p}{12} - \left(\frac{p}{4}\right)^2 = \frac{p \cdot (4 - 3 \cdot p)}{48}. \end{aligned} \tag{S20}$$

To obtain the population intercept and correlation, we also need the covariance between X and U, which is defined as

$$\text{Cov}(X, U) = \mathbb{E}(X \cdot U) - \mathbb{E}(X) \cdot \mathbb{E}(U). \tag{S21}$$

Since we already know the population means (Equation S17), we only need to calculate $\mathbb{E}(X \cdot U)$. As both variables are zero for unaware participants, this simplifies to

$$\mathbb{E}(X \cdot U) = p \cdot \mathbb{E}(X \cdot U \mid \text{aware}). \tag{S22}$$

For aware participants, $X = 2 \cdot |Z|$ and $U = \Phi(|Z|) - .5$, where Z is a standard normal variable. Since the density of $|Z|$ is $2 \cdot \phi(z)$ for $z \geq 0$, their expected product is

$$\begin{aligned} \mathbb{E}(X \cdot U \mid \text{aware}) &= 4 \int_0^\infty z \cdot [\Phi(z) - .5] \cdot \phi(z)\, dz \\ &= 4 \int_0^\infty \phi(z)^2\, dz \\ &= \frac{1}{\sqrt{\pi}}, \end{aligned} \tag{S23}$$

where the second equality follows from integration by parts, using $\phi'(z) = -z \cdot \phi(z)$. Therefore, substituting the population means (Equation S17) and the expected product (Equation S22 and Equation S23) into Equation S21 gives

$$\begin{aligned} \text{Cov}(X, U) &= \frac{p}{\sqrt{\pi}} - \left(2p\sqrt{\frac{2}{\pi}}\right) \cdot \frac{p}{4} \\ &= \frac{p}{\sqrt{\pi}} \cdot \left(1 - \frac{p}{\sqrt{2}}\right). \end{aligned} \tag{S24}$$

We can now use these population moments (Equation S17, Equation S20, and Equation S24) to derive the regression intercept. Recall that, under the Goldstein model, influence-task sensitivity is generated as $d'_{iI} = \alpha + \beta \cdot U + \varepsilon_i$ (Equation S9), where ε_i is independent of X and U and has mean zero. Therefore, $\mu_I = \alpha + \beta \cdot \mathbb{E}(U)$ and $\text{Cov}(X, d'_{iI}) = \beta \cdot \text{Cov}(X, U)$. Substituting these expressions into Equation S10 gives

$$\begin{aligned}
\beta_0 - \alpha &= \beta \cdot \left[\mathbb{E}(U) - \frac{\text{Cov}(X, U)}{\mathbb{V}(X)} \cdot \mathbb{E}(X) \right] \\
&= \beta \cdot \left[\frac{p}{4} - \frac{p \cdot (1 - p/\sqrt{2})/\sqrt{\pi}}{4 \cdot p \cdot (1 - 2 \cdot p/\pi)} \cdot 2 \cdot p \cdot \sqrt{\frac{2}{\pi}} \right] \\
&= \beta \cdot \frac{p \cdot (\pi - 2 \cdot \sqrt{2})}{4 \cdot (\pi - 2 \cdot p)}.
\end{aligned} \tag{S25}$$

Because $\pi > 2 \cdot \sqrt{2}$ and $0 < p \leq 1$, the difference $\beta_0 - \alpha$ is positive whenever $\beta > 0$. Thus, although α represents the expected influence-task sensitivity of unaware participants, β_0 is the intercept of the best linear approximation to the relationship expressed on the d' scale.

Finally, we derive the correlation between true sensitivities. The generating correlation ρ is defined between U and d'_{iI}, whereas the multilevel SDT model describes the correlation between X and d'_{iI}. Since the residuals are independent of both awareness measures,

$$\begin{aligned}
\rho_d &= \frac{\text{Cov}(X, d'_{iI})}{\sqrt{\mathbb{V}(X)} \cdot \sigma_I} \\
&= \frac{\beta \cdot \text{Cov}(X, U)}{\sqrt{\mathbb{V}(X)} \cdot \sigma_I} \\
&= \underbrace{\frac{\beta \cdot \sqrt{\mathbb{V}(U)}}{\sigma_I}}_{\rho} \cdot \underbrace{\frac{\text{Cov}(X, U)}{\sqrt{\mathbb{V}(X) \cdot \mathbb{V}(U)}}}_{\text{Cor}(X,U)} \\
&= \rho \cdot \frac{\sqrt{12/\pi} \cdot (1 - p/\sqrt{2})}{\sqrt{(1 - 2 \cdot p/\pi) \cdot (4 - 3 \cdot p)}}.
\end{aligned} \tag{S26}$$

Thus, the correlation between true sensitivities differs from the generating correlation because awareness accuracy and sensitivity are related nonlinearly. We therefore distinguished the

generating intercept α from the population linear-projection intercept β_0 when evaluating the results. The resulting population values are reported in Table S8.

Table S8
Population Values Under the Generating Models of Simulation Study 2

			ρ_d			$\beta_0 - \alpha$		
π_0	μ_A	σ_A	$\rho = .2$	$\rho = .4$	$\rho = .6$	$\rho = .2$	$\rho = .4$	$\rho = .6$
Klauer et al. (1998)								
.2	1.91	1.67	.200	.400	.600	0.000	0.000	0.000
.5	0.80	1.17	.200	.400	.600	0.000	0.000	0.000
.8	0.22	0.61	.200	.400	.600	0.000	0.000	0.000
Goldstein et al. (2022)								
.2	1.28	1.25	.192	.383	.575	0.015	0.030	0.045
.5	0.80	1.17	.194	.387	.581	0.007	0.014	0.020
.8	0.32	0.84	.195	.390	.584	0.003	0.006	0.009

Note. π_0 = proportion of unaware participants; μ_A and σ_A = mean and SD of true awareness-task sensitivity; ρ_d = correlation between true sensitivities; $\beta_0 - \alpha$ = difference between the intercept of the multilevel model and the intercept of the generating model. Values for a generating correlation ρ of .2, .4, or .6, with $\sigma_I = 0.3$; they do not depend on α.

Reliability of the Observed Sensitivities

Reliability was not a design factor in Study 2. The SDs of true sensitivities were fixed by each model ($\sigma_I = 0.3$, and σ_A as in Table S8), so the reliability of the observed sensitivities depended on the number of trials and, in the awareness task, on the proportion of unaware participants. Equation S5 cannot be applied to the awareness task, because its expected sampling variance is infinite: for near-perfect participants, the sampling variance of $\hat{d}'$ grows faster than the density of d'_{iA} decreases. We therefore computed reliability as the squared correlation between true and observed sensitivities in 100,000 participants simulated as in each condition, with observed sensitivities computed with the log-linear correction (Hautus, 1995) and, in the model of Goldstein et al., after the median split of the RTs. In the conditions of Study 1, this measure differed from the nominal reliability by at most .008.

Table S9 shows the results. The reliability of the influence task depended only on the number of trials (.22, .46, and .58 with 10, 30, and 50 trials, varying by at most .02 across models,

unaware proportions, correlations, and intercepts), comparable to reliabilities of .2, .5, and .6 in Study 1. The awareness task was more reliable (.48 to .93), because true awareness-task sensitivities were much more variable than influence-task sensitivities, and it was least reliable with 80% of unaware participants, when σ_A was smallest. Unlike Study 1, in which both tasks had the same reliability, Study 2 thus combined a reliable awareness task with an influence task of low to moderate reliability.

Table S9
Reliability of the Observed Sensitivities in Simulation Study 2

	Awareness task			Influence task		
π_0	K = 10	K = 30	K = 50	K = 10	K = 30	K = 50
Klauer et al. (1998)						
.2	.77	.90	.93	.22	.46	.58
.5	.73	.89	.93	.22	.46	.59
.8	.48	.75	.84	.22	.46	.59
Goldstein et al. (2022)						
.2	.73	.89	.93	.21	.45	.58
.5	.73	.89	.93	.22	.46	.58
.8	.60	.83	.90	.22	.46	.58

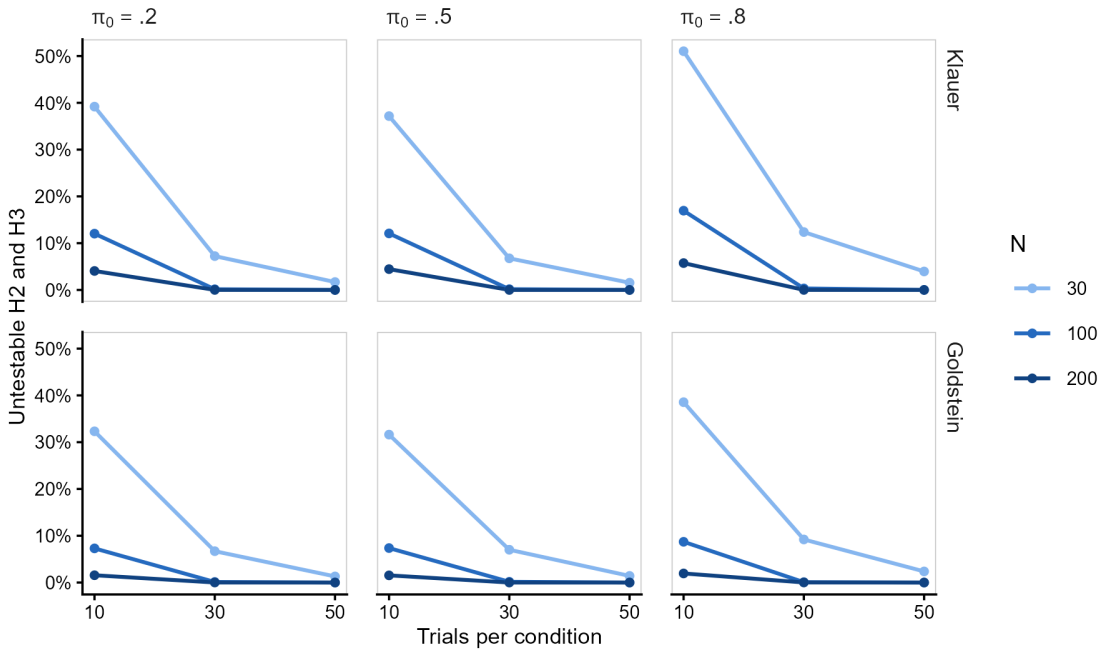
Note. Squared correlation between true and observed sensitivities in 100,000 simulated participants per condition, averaged over the generating correlations and intercepts. π_0 = proportion of unaware participants; K = trials per condition and task.

Convergence and Boundary Fits

All models converged except 19 fits under the model of Klauer et al. and 12 under the model of Goldstein et al. (out of 324,000 each). Untestable fits were less frequent than in Study 1 (8% and 6% on average, with maxima of 57% and 46%), because the awareness task was more reliable in both models (Table S9). As Figure S9 shows, they occurred mostly with 10 trials per condition, when the influence task was least reliable (20% and 15% on average, compared with 0.8% and 0.6% with 50 trials), and with 30 participants. They were somewhat more frequent with 80% of unaware participants, when the awareness task was least variable.

Figure S9

Untestable Fits by Trials, Sample Size, and Proportion of Unaware Participants in Simulation Study 2



Note. Proportion of fits without SEs for the correlation and the intercept, averaged over generating correlations and intercepts.

Recovery and Inference

Table S10 summarizes bias, coverage, and rejection rates under the null hypothesis for the correlation and the intercept, and Figure S10 shows bias and coverage by trials and sample size. The correlation was slightly overestimated with few trials, and its intervals were conservative with 30 participants or 10 trials; its Type I error rate never exceeded .08 and was conservative in 14% of the null conditions. The intercept of the multilevel model was recovered with little bias, its coverage was close to .95, and its Type I error rate was within Bradley's criterion in nearly every null condition. Against the generating intercept α, the estimates of Goldstein's model were positively biased, coverage fell to .827 in the worst condition, and the rejection rate when $\alpha = 0$ exceeded .075 in 13% of the null conditions (see the last section).

Table S10

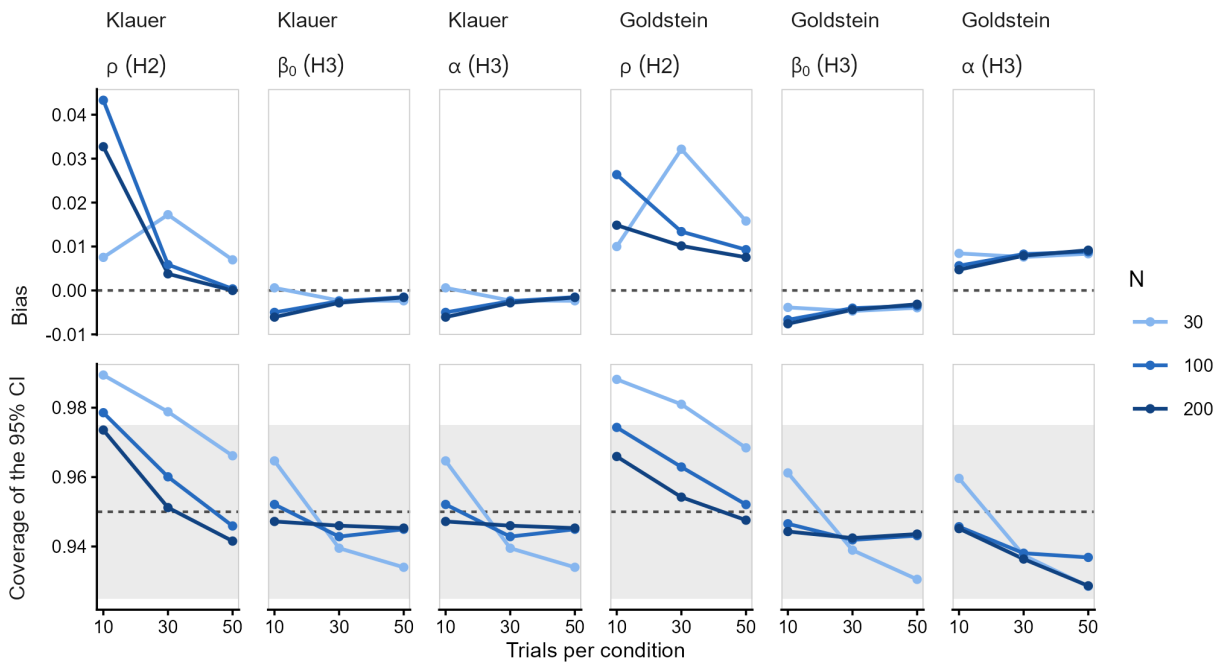
Bias, Coverage, and Rejection Rates Under the Null Hypothesis in Simulation Study 2

		ECR			Rejection under H0		
Target	Bias	Mean [range]	Liberal	Conservative	Mean [max]	Liberal	Conservative
Klauer et al. (1998)							
H2: Correlation	0.013 [−0.114, 0.087]	0.965 [0.904, 1.000]	2%	31%	0.040 [0.063]	0%	14%
H3: Intercept (β_0)	−0.003 [−0.020, 0.022]	0.946 [0.903, 0.985]	2%	2%	0.054 [0.075]	1%	2%
H3: Intercept (α)	−0.003 [−0.020, 0.022]	0.946 [0.903, 0.985]	2%	2%	0.054 [0.075]	1%	2%
Goldstein et al. (2022)							
H2: Correlation	0.016 [−0.041, 0.081]	0.966 [0.927, 1.000]	0%	32%	0.040 [0.076]	1%	14%
H3: Intercept (β_0)	−0.005 [−0.031, 0.017]	0.944 [0.912, 0.976]	5%	1%	0.057 [0.080]	4%	0%
H3: Intercept (α)	0.008 [−0.026, 0.041]	0.940 [0.827, 0.975]	13%	0%	0.060 [0.151]	13%	0%

Note. Mean [minimum, maximum] over conditions for bias and ECR, and mean [maximum] over the null conditions for the rejection rate over testable fits. The liberal and conservative columns are the percentages of conditions outside Bradley's liberal criterion. β_0 = intercept of the multilevel model; α = intercept of the generating model (equal to β_0 in the model of Klauer et al.). ECR = empirical coverage rate of the 95% CIs.

Figure S10

Bias and Coverage of the Correlation and the Intercept by Trials and Sample Size in Simulation Study 2



Note. Means over the proportion of unaware participants and the generating correlations and intercepts. β_0 = intercept of the multilevel model; α = intercept of the generating model. The gray band in the coverage panels shows Bradley's liberal criterion, [.925, .975].

Meta-Models

Table S11 and Table S12 report the meta-models of each generating model, and Table S13 and Table S14 the marginal means. In both models, the RMSE of both parameters depended on sample size and trials, and the proportion of untestable fits on trials, sample size, and their interaction (ω^2 = .41, .27, and .23 under the model of Klauer et al.; .33, .33, and .28 under the model of Goldstein et al.). The coverage of the correlation depended on sample size and trials, because its intervals were conservative with few participants or few trials. Its bias depended on the number of trials under the model of Klauer et al. (ω^2 = .16) and on the generating correlation under the model of Goldstein et al. (ω^2 = .30), but remained small in absolute terms (Table S10). The bias of the intercept depended on the generating correlation (ω^2 = .20 and .38): the stronger the correlation, the more the intercept was underestimated, by at most .03. Its coverage decreased slightly with more trials and larger samples (trials: ω^2 = .26 and .20), as this small bias became larger relative to the SEs.

Table S11

Effect Sizes of the Meta-Models Under the Model of Klauer et al. (1998) in Simulation Study 2

	H2: ρ			H3: β_0			Boundary
Predictor	Bias	RMSE	ECR	Bias	RMSE	ECR	UNT
N	0.01	**0.48**	**0.25**	0.02	**0.48**	0.00	**0.27**
K	**0.16**	**0.41**	**0.39**	0.01	**0.24**	**0.26**	**0.41**
π_0	0.08	0.03	0.00	0.10	0.03	0.04	0.01
ρ	0.08	0.02	0.04	**0.20**	0.00	0.00	0.02
$N \times K$	0.11	0.04	0.01	0.06	0.08	**0.21**	**0.23**
$K \times \rho$	0.05	0.00	0.07	0.00	0.00	0.00	0.02
$\pi_0 \times \rho$	0.07	0.00	0.03	0.04	0.00	0.00	0.00
$N \times K \times \rho$	0.10	0.00	0.02	0.01	0.00	0.01	0.00

Note. Non-partial ω^2. N = sample size; K = trials per condition; π_0 = proportion of unaware participants; ρ = generating correlation; α = generating intercept. H3 is evaluated against the intercept of the multilevel model, β_0. RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs; UNT = untestable fits for H2 and H3. Effects with $\omega^2 < .06$ in every measure are omitted, and values of at least .14 are in bold.

Table S12

Effect Sizes of the Meta-Models Under the Model of Goldstein et al. (2022) in Simulation Study 2

	H2: ρ			H3: β_0			Boundary
Predictor	Bias	RMSE	ECR	Bias	RMSE	ECR	UNT
N	0.03	**0.55**	**0.34**	0.00	**0.46**	0.00	**0.33**
K	0.03	**0.34**	**0.24**	0.02	**0.25**	**0.20**	**0.33**
π_0	0.04	0.01	0.00	0.08	0.02	0.02	0.00
ρ	**0.30**	0.02	0.01	**0.38**	0.00	0.01	0.02
$N \times K$	0.11	0.06	0.00	0.01	0.08	**0.25**	**0.28**
$K \times \rho$	0.03	0.00	**0.19**	0.02	0.00	0.00	0.01
$\pi_0 \times \rho$	0.04	0.00	0.02	0.06	0.00	0.01	0.00

Note. Non-partial ω^2. Abbreviations as in the previous table. H3 is evaluated against the intercept of the multilevel model, β_0. Effects with $\omega^2 < .06$ in every measure are omitted, and values of at least .14 are in bold.

Intercept of the Generating Model

Figure S11 shows the rejection rate of H3 when the generating intercept is zero ($\alpha = 0$). Under the model of Klauer et al., the intercept of the multilevel model equals α, and the rejection rate stayed close to .05 in every condition. Under the model of Goldstein et al., the intercept of the multilevel model exceeds α by $\beta \cdot \kappa(p)$ (Equation S25), so H3 tests a hypothesis that is false whenever the generating correlation is positive. The consequence was negligible with few aware participants or weak correlations, but with 80% of aware participants ($\pi_0 = .2$) and a generating correlation of .6, the rejection rate grew with the sample size and the number of trials, reaching 15% with $N = 200$ and $K = 50$. This rejection rate is not an error of the estimator, which recovered β_0 accurately, but a consequence of the definition of the intercept as the best linear predictor. It would increase further with larger samples, because the estimate converges to β_0, not to α, and the same would happen with any method that relies on a linear regression of influence-task sensitivity on awareness-task d'.

Table S13

Marginal Means Under the Model of Klauer et al. (1998) in Simulation Study 2

	H2: ρ			H3: β_0			Boundary
Level	Bias	RMSE	ECR	Bias	RMSE	ECR	UNT
Sample size (N)							
30	0.011	0.428	**0.978**	−0.001	0.134	0.946	0.179
100	0.017	0.225	0.962	−0.003	0.067	0.947	0.046
200	0.012	0.151	0.955	−0.003	0.047	0.946	0.016
Trials per condition (K)							
10	0.028	0.419	**0.981**	−0.003	0.120	0.955	0.203
30	0.009	0.216	0.963	−0.003	0.069	0.943	0.030
50	0.002	0.170	0.951	−0.002	0.059	0.941	0.008
Unaware participants (π_0)							
0.2	0.021	0.245	0.966	−0.005	0.095	0.944	0.072
0.5	0.016	0.250	0.965	−0.002	0.074	0.945	0.069
0.8	0.003	0.310	0.964	−0.001	0.079	0.950	0.100
Generating correlation (ρ)							
0	−0.000	0.291	0.969	0.001	0.084	0.945	0.062
0.2	0.015	0.283	0.968	−0.002	0.081	0.946	0.065
0.4	0.020	0.266	0.964	−0.004	0.087	0.946	0.079
0.6	0.017	0.234	0.959	−0.005	0.078	0.948	0.115
Generating intercept (α)							
0	0.013	0.267	0.964	−0.003	0.081	0.946	0.079
0.15	0.013	0.269	0.965	−0.003	0.085	0.947	0.081
0.3	0.014	0.268	0.966	−0.002	0.082	0.946	0.081

Note. Means over the remaining factors. H3 is evaluated against the intercept of the multilevel model, β_0. RMSE = root mean squared error; ECR = empirical coverage rate of the 95% CIs; UNT = untestable fits for H2 and H3. Coverages outside [.925, .975] are in bold.

Table S14

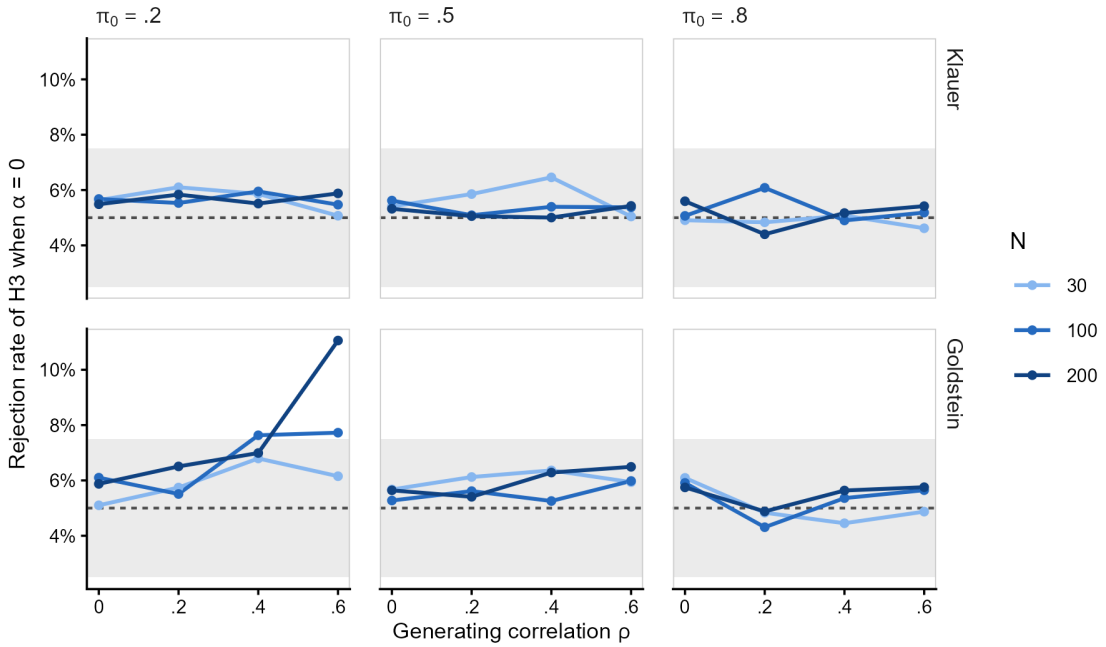
Marginal Means Under the Model of Goldstein et al. (2022) in Simulation Study 2

	H2: ρ			H3: β_0			Boundary
Level	Bias	RMSE	ECR	Bias	RMSE	ECR	UNT
Sample size (N)							
30	0.019	0.401	**0.979**	−0.004	0.130	0.944	0.145
100	0.016	0.201	0.963	−0.005	0.066	0.944	0.026
200	0.011	0.133	0.956	−0.005	0.047	0.943	0.006
Trials per condition (K)							
10	0.017	0.369	**0.976**	−0.006	0.118	0.951	0.145
30	0.019	0.205	0.966	−0.004	0.067	0.941	0.026
50	0.011	0.161	0.956	−0.004	0.058	0.939	0.006
Unaware participants (π_0)							
0.2	0.019	0.235	0.967	−0.007	0.093	0.942	0.055
0.5	0.017	0.236	0.966	−0.004	0.075	0.943	0.055
0.8	0.010	0.265	0.965	−0.002	0.075	0.946	0.068
Generating correlation (ρ)							
0	−0.000	0.266	0.968	0.001	0.081	0.943	0.045
0.2	0.014	0.259	0.967	−0.003	0.081	0.946	0.047
0.4	0.023	0.243	0.965	−0.007	0.085	0.943	0.059
0.6	0.025	0.213	0.965	−0.010	0.077	0.942	0.085
Generating intercept (α)							
0	0.017	0.245	0.966	−0.006	0.081	0.943	0.059
0.15	0.015	0.245	0.966	−0.005	0.082	0.943	0.059
0.3	0.015	0.245	0.966	−0.003	0.080	0.945	0.059

Note. Means over the remaining factors. Abbreviations as in the previous table. Coverages outside [.925, .975] are in bold.

Figure S11

Rejection Rate of H3 When the Intercept of the Generating Model Is Zero in Simulation Study 2



Note. Rejection rates over testable fits in the conditions with $\alpha = 0$, averaged over trials per condition. The gray band shows Bradley's liberal criterion, [.025, .075], and the dashed line the nominal .05.

References

Bradley, J. V. (1978). Robustness? *British Journal of Mathematical and Statistical Psychology*, *31*(2), 144–152. https://doi.org/10.1111/j.2044-8317.1978.tb00581.x

Chalmers, R. P., & Adkins, M. C. (2020). Writing effective and reliable Monte Carlo simulations with the SimDesign package. *The Quantitative Methods for Psychology*, *16*(4), 248–280. https://doi.org/10.20982/tqmp.16.4.p248

Cohen, J. (1992). Statistical power analysis. *Current Directions in Psychological Science*, *1*(3), 98–101. https://doi.org/10.1111/1467-8721.ep10768783

Goldstein, A., Sklar, A. Y., & Siegelman, N. (2022). Accurately measuring nonconscious processing using a generative Bayesian framework. *Psychology of Consciousness: Theory, Research, and Practice*, *9*(4), 336–355. https://doi.org/10.1037/cns0000316

Gourevitch, V., & Galanter, E. (1967). A significance test for one parameter isosensitivity functions. *Psychometrika*, *32*(1), 25–33. https://doi.org/10.1007/BF02289402

Greene, W. H. (2011). *Econometric analysis* (7th ed.). Prentice Hall.

Hautus, M. J. (1995). Corrections for extreme proportions and their biasing effects on estimated values of d'. *Behavior Research Methods, Instruments, & Computers*, *27*(1), 46–51. https://doi.org/10.3758/BF03203619

Klauer, K. C., Draine, S. C., & Greenwald, A. G. (1998). An unbiased errors-in-variables approach to detecting unconscious cognition. *British Journal of Mathematical and Statistical Psychology*, *51*(2), 253–267. https://doi.org/10.1111/j.2044-8317.1998.tb00680.x

Lin, L. I.-K. (1989). A concordance correlation coefficient to evaluate reproducibility. *Biometrics*, *45*(1), 255–268. https://doi.org/10.2307/2532051

Meyen, S., Zerweck, I. A., Amado, C., von Luxburg, U., & Franz, V. H. (2022). Advancing research on unconscious priming: When can scientists claim an indirect task advantage? *Journal of Experimental Psychology: General*, *151*(1), 65–81. https://doi.org/10.1037/xge0001065

Rey-Sáez, R. (2026). *uSDT_pub*. GitHub repository. https://github.com/RicardoReySaez/uSDT_pub

Rey-Sáez, R., Franco-Martínez, A., Vadillo, M., & Garre-Frutos, F. (2026). *Code, data, and materials for: A multilevel signal detection theory approach to unconscious processing research* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.24019945

Skrondal, A. (2000). Design and analysis of Monte Carlo experiments: Attacking the conventional wisdom. *Multivariate Behavioral Research*, *35*(2), 137–167. https://doi.org/10.1207/S15327906MBR3502_1

Suero, M., Privado, J., & Botella, J. (2017). Methods to estimate the variance of some indices of the signal detection theory: A simulation study. *Psicológica*, *38*(1), 149–175.